Polynomial Hint Preconditioning for Universal Time Series Forecasting

Anonymous

Abstract

Patch-based transformer architectures for time series forecasting segment the input into fixed-length patches, creating an information bottleneck: each patch embedding is initially blind to temporal patterns spanning multiple patches. We propose **polynomial hint preconditioning**, a method that applies a fixed Chebyshev polynomial FIR filter to the raw time series and concatenates the filtered signal as an auxiliary input channel, injecting cross-patch temporal information with negligible overhead (12K parameters, 0.11% of the model). On the GIFT-Eval benchmark (97 dataset×horizon configurations, 5 random seeds), our method achieves a 2.9% improvement in geometric mean MASE over the baseline ($p < 10^{-15}$, paired sign test), winning 329 of 485 comparisons. Controlled capacity ablations confirm the improvement stems from the *polynomial content*: a duplicate input channel *hurts* (44.7% win rate), while a zero channel is neutral. At 100K training steps, the benefit grows to 3.9%. The benefit is strongest for high-frequency domains and long prediction horizons, consistent with cross-patch information leakage at the filter's lookback scale.

1 Introduction

Universal time series forecasting models [Woo et al., 2024, Das et al., 2024, Ansari et al., 2024, Rasul et al., 2023, Goswami et al., 2024] train a single model on diverse datasets and generalize zero-shot to unseen domains [Liang et al., 2024]. The Moirai family [Woo et al., 2024, Liu et al., 2024] is among the most successful, using a patch-based transformer encoder where the input is split into non-overlapping patches of size P (typically 16), each projected into a d-dimensional embedding—inspired by Vision Transformers [Dosovitskiy et al., 2021] and PatchTST [Nie et al., 2023].

This introduces a *cross-patch information bottleneck*. Before self-attention, each patch embedding encodes only P consecutive time steps. Patterns spanning multiple patches—daily cycles in hourly data (24 steps = 1.5 patches), weekly seasonality in 15-minute data (672 steps = 42 patches)—must be recovered entirely through attention. PatchTST [Nie et al., 2023] addresses this with overlapping patches (increasing sequence length), and Pathformer [Chen et al., 2024] uses adaptive multi-scale patches, but the fundamental bottleneck remains.

We propose **polynomial hint preconditioning**: before patching, we apply a fixed Chebyshev polynomial FIR filter [Oppenheim et al., 1999] to the raw time series and concatenate the filtered signal as an additional channel:

$$h[t] = T_d(B^s)\, x[t], \quad s = P, \tag{1}$$

where T_d is the Chebyshev polynomial of degree d and B^s is the backshift operator. For d=4, s=16: $h[t] = x[t]-2x[t-32]+0.5x[t-64]$ injecting information from 2 and 4 patches back. Only the input projection widens (12K extra parameters out of 11.4M, +0.11%).

Our approach is inspired by *universal sequence preconditioning* [Marsden and Hazan, 2024], which shows polynomial transformations reduce spectral complexity of sequence prediction. We adapt this insight as an auxiliary "hint" channel rather than replacing the original input.

Contributions.

1. We introduce polynomial hint preconditioning, adding a fixed Chebyshev FIR-filtered channel to patch-based time series transformers.
2. Through 5-seed experiments on 97 GIFT-Eval configurations, we show 2.9% improvement ($p < 10^{-15}$) with capacity ablations confirming polynomial content drives the gain. At 100K steps, the benefit grows to 3.9%.
3. Extensive ablations identify optimal settings: Chebyshev basis, degree 4, patch-aligned stride, 10% hint dropout.
4. Domain/frequency analysis shows the benefit is strongest where the filter's lookback aligns with structured temporal patterns.

2 Related Work

Time series foundation models. Recent work trains large models on diverse corpora for zero-shot forecasting [Liang et al., 2024]. Approaches include decoder-only models (TimesFM [Das et al., 2024], Lag-Llama [Rasul et al., 2023]), tokenization-based (Chronos [Ansari et al., 2024]), and masked encoders (Moirai [Woo et al., 2024], Moirai-2 [Liu et al., 2024], MOMENT [Goswami et al., 2024], TTM [Ekambaram et al., 2024]). We build on Moirai-2 Small (11.4M parameters).

Preconditioning for sequences. Marsden and Hazan [2024] show Chebyshev polynomial transformations provide near-optimal spectral conditioning, building on Hazan et al. [2017, 2018]. Recent spectral SSMs [Agarwal et al., 2023, Dao et al., 2024] also leverage polynomial structure. Our work bridges this theory with practical transformers.

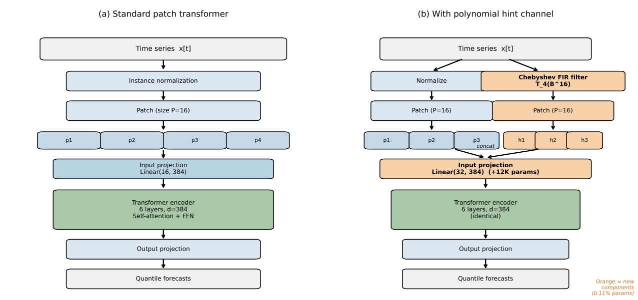

Figure 1: Overview. (a) Standard patch transformer. (b) Our method adds a Chebyshev FIR-filtered hint channel (orange) concatenated before the input projection. Only the input projection is wider; the transformer encoder is unchanged. Overhead: 12K parameters (0.11%).

Patch-based models. PatchTST [Nie et al., 2023] introduced channel-independent patching; Moirai [Woo et al., 2024] adds any-variate attention; Han et al. [2024] analyze the capacity–robustness tradeoff. Our approach is orthogonal: we inject cross-patch information at the input level through a fixed polynomial filter, requiring no changes to attention or patching.

3 Method

Background. Following Moirai [Woo et al., 2024, Liu et al., 2024], a normalized time series is partitioned into non-overlapping patches of size P. Each patch $\mathbf{p}_i$ and its mask $\mathbf{m}_i$ form a $2P$-dimensional vector projected to d_{model} via a residual block: $\mathbf{e}_i = \text{ResBlock}_{2P\to d}([\mathbf{p}_i; \mathbf{m}_i])$.

Chebyshev FIR filter. We compute $h[t] = T_d(B^s)x[t] = \sum_{k=0}^{d} c_k x[t-ks]$, where $c_0, \dots, c_d$ are fixed Chebyshev coefficients. For d=4: $T_4(z) = 8z^4 - 8z^2 + 1$.

Hint channel. The hint residual $\tilde{h}[t] = h[t] - x[t]$ is computed, masked, and patched. The input becomes $3P$-dimensional:

$$\mathbf{e}_i = \text{ResBlock}_{3P\to d}([\mathbf{p}_i; \mathbf{m}_i; \tilde{\mathbf{h}}_i]). \tag{2}$$

Hint dropout. During training, each patch's hint is zeroed with probability p_{drop}=0.1, preventing over-reliance. Full hints are used at inference.

Parameter overhead. Only the first linear layer and residual skip of the ResBlock widen ($3P$ vs. $2P$ inputs). Total: +12,288 parameters (+0.11%) for Moirai-2 Small.

4 Experimental Setup

Base model. Moirai-2 Small [Liu et al., 2024]: 6-layer transformer, d=384, patch size P=16, 11.4M parameters. Retrained from scratch on LOTSA v1 [Woo et al., 2024] with Moirai-2's weighted variate-proportional sampling.

Table 1: Main results (5 seeds, 485 comparisons). "Wins" = configs with strictly lower MASE than Baseline. p-values from binomial sign test.

Method	MASE	vs BL	Wins	p
HD10 (Ours)	**0.837**	**−2.9%**	**329/485**	$\mathbf{10^{-15}}$
Zero channel	0.858	−0.5%	262/485	0.042
Baseline	0.862	—	—	—
Duplicate	0.865	+0.4%	217/485	0.99

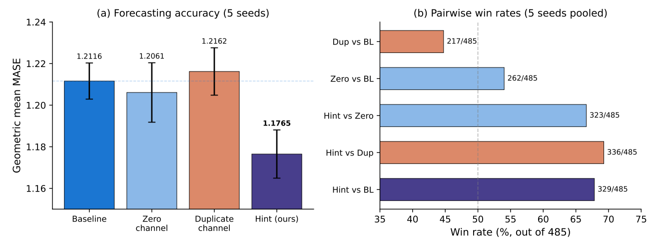

Figure 2: **Main result.** (a) Geometric mean MASE across 5 seeds (bars: std). HD10 significantly outperforms baseline and capacity controls. Duplicate *hurts*. (b) Pooled win rates.

Training. 10K steps (also 100K for extended runs), batch size 256, AdamW (β_1=0.9, β_2=0.98, wd =0.1), cosine LR with 1K warmup, peak η=10^{-3}, bf16.

Evaluation. GIFT-Eval benchmark [Vijay et al., 2024]: 97 dataset×horizon configs, 9 domains, 7 frequencies. Metric: geometric mean MASE at context length 4,000. We report normalized MASE following the GIFT-Eval leaderboard convention (dividing by the seasonal naive MASE). Statistical test: paired sign test pooled across 5 seeds ($5 \times 97 = 485$ comparisons).

Conditions. **Baseline**: standard Moirai-2 Small. **Zero**: hint architecture with all-zeros hint. **Duplicate**: hint architecture with copy of input as hint. **HD10 (Ours)**: Chebyshev d=4, stride 16, 10% dropout. Zero and Duplicate are capacity controls with identical architecture/parameter count.

5 Results

5.1 Main Result

Table 1 shows the main results. HD10 achieves 0.837 MASE, a 2.9% improvement winning 329/485 comparisons ($p < 10^{-15}$). The capacity controls confirm polynomial content drives the gain: Duplicate *hurts* (44.7% wins), Zero is neutral (54.0%), and HD10 beats both directly (vs. Duplicate: 336/485, $p = 6 \times 10^{-18}$; vs. Zero: 323/485, $p = 10^{-13}$).

5.2 Matched Learning Rate

The baseline is LR-sensitive: increasing to $\eta = 2 \times 10^{-3}$ improves it by 2.3% (to 0.842). At this matched LR, HD10

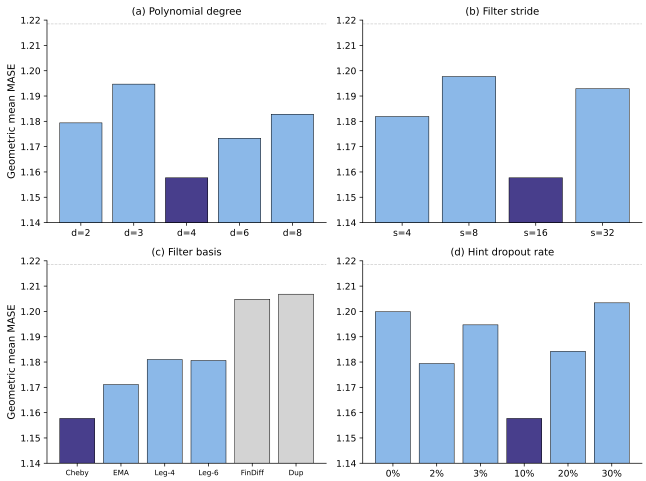

Figure 3: **Ablations** (seed 0, 97 configs). Numbers: wins/97 vs. baseline. Orange = optimal. (a) Degree: d=4 best. (b) Stride: s=16=P best. (c) Basis: Chebyshev best. (d) Dropout: 10% best.

still wins (280/485, $p = 3.8 \times 10^{-4}$) but the effect is smaller (+0.7%). Crucially, HD10 is *LR-insensitive*: 0.837 at η=10^{-3} vs. 0.836 at η=$2{\times}10^{-3}$ (0.1% difference), suggesting hints stabilize optimization across learning rates.

5.3 Ablations

Figure 3 summarizes ablations (seed 0, 10% dropout unless noted):

Degree. d=4 is optimal (0.823, -5.0%). Even degrees (d=2, 4, 6, 8) outperform odd (d=3, 5, 7).

Stride. s=16=P is optimal. Patch-aligned stride provides maximally informative cross-patch lookback.

Basis. Chebyshev (-5.0%) > EMA (-3.9%) > Legendre (-3.1%) > Finite Diff (-1.1%).

Dropout. Inverted-U: 10% optimal. Without dropout, the model becomes over-dependent (35.9% degradation when hints removed at inference vs. 9.3% with 10% dropout).

Inference ablation. Replacing the Chebyshev hint with alternatives at inference confirms the model learns specific polynomial structure: Normal $\gg$ Baseline (+5.2%) > Zeros (+9.3%) > Random (+14.5%) > Duplicate (+29.9%).

5.4 Domain and Frequency Analysis

Figure 4 shows the benefit is heterogeneous. Strongest: Transport (+6.9%, $p = 4 \times 10^{-8}$), Energy (+4.0%, $p = 2 \times 10^{-4}$), Demographics (+9.8%). Neutral: Weather (+0.3%, ns), Healthcare (+0.2%, ns). This is consistent with the mechanism: at stride s=16, the filter looks back 32 and 64 steps.

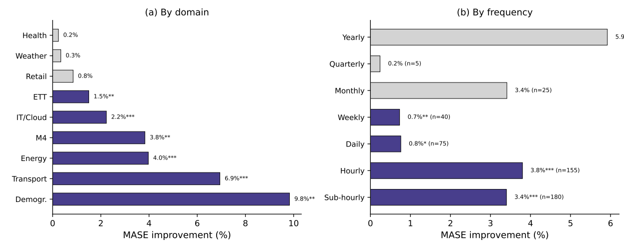

Figure 4: **Domain and frequency analysis** (5 seeds pooled). Green: $p < 0.05$. Gray: not significant. Strongest in transport, energy, and sub-hourly/hourly.

Table 2: 100K training budget (5 seeds). HD10 outperforms baseline by 3.9% and capacity controls by >4.5%. Both Duplicate and Zero *hurt* at 100K.

	s0	s1	s2	s7	s42	Mean
HD10	**0.835**	**0.840**	0.858	**0.845**	**0.843**	**0.844**
BL	0.909	0.892	**0.863**	0.864	0.863	0.878
Dup	0.889	0.874	0.898	0.879	0.890	0.886
Zero	0.892	0.858	0.875	0.883	0.961	0.894

For 15-minute data, this is 8 and 16 hours (intra-day patterns). For daily weather, 32–64 days is misaligned with relevant patterns.

The benefit also increases with prediction horizon: $+1.4\%$ for short horizons to $+5.1\%$ for long horizons. Harder configurations (high baseline MASE) benefit more: $+5.9\%$ for the hardest quartile vs. $+0.4\%$ for the easiest.

5.5 Extended Training (100K Steps)

Table 2 shows 100K-step results. The hint benefit *grows* with training budget: HD10 improves 3.9% over baseline (vs. 2.9% at 10K). Capacity controls that were near-neutral at 10K now actively hurt: Duplicate -0.9%, Zero -1.7%. This suggests uninformative channels introduce optimization difficulties that compound, while polynomial hints provide compounding inductive bias. The baseline shows higher cross-seed variance (seed 0 reaches 0.909 due to late instability), while HD10 remains stable (range 0.83–0.86).

5.6 Additional Validation

We confirm generalization on FEV-Bench [Vijay et al., 2024] (100 tasks): HD10 wins 72/100 configurations with a 2.5% MASE improvement (Table 8). Under the official Moirai-2 protocol (10K warmup, 100K steps), HD10 still improves by 2.3% (Table 9).

5.7 Mechanism

The hint channel injects cross-patch information at the input level. For Chebyshev d=4, s=16: $\tilde{h}[t] = -2x[t{-}32] + 0.5x[t{-}64]$, providing each patch with weighted values from patches 2 and 4 steps back. This reduces the burden on self-attention to discover these dependencies.

HD10 also reduces per-configuration variance by 10.7% relative to baseline. Configurations with high baseline variance benefit most (-6.6% improvement for highest-variance quartile vs. -0.4% for lowest), suggesting hints act as a targeted regularizer. Training loss analysis confirms this: at 10K steps, HD10 achieves only 1.0% lower training loss (0.1350 vs. 0.1364) but 2.9% lower eval MASE—a $2.9\times$ amplification consistent with a regularization rather than optimization effect. At 100K steps, the gap widens (2.4% train vs. 3.9% eval), and the baseline's eval instability at 80K is not reflected in its training loss.

6 Limitations

LR interaction: The improvement at default LR (-2.9%) shrinks at matched optimal LR (-0.7%); part of the benefit is optimization stability. **Domain heterogeneity**: No significant improvement on weather/healthcare where the filter's lookback misaligns with relevant temporal scales. **Single scale**: Preliminary 46M-parameter experiments show inconsistent results, suggesting the benefit is specific to capacity-constrained models. **Fixed coefficients**: Learned FIR coefficients are competitive but slightly worse than fixed Chebyshev, suggesting the polynomial structure is near-optimal.

7 Conclusion

We showed that a fixed polynomial FIR filter applied as an auxiliary input channel to a patch-based time series transformer provides consistent improvement across seeds, domains, and training budgets. The method adds negligible overhead (0.11% parameters, $\sim$1.6% compute) and requires no modification to the transformer encoder, making it applicable to any patch-based architecture. Through controlled ablations with duplicate and zero channels, we confirm the improvement is driven by polynomial content, not extra capacity. Our work provides empirical evidence for the practical value of polynomial basis transformations [Marsden and Hazan, 2024, Hazan et al., 2017] in time series foundation models [Liang et al., 2024].

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Supplementary Material

A Detailed Experimental Methodology

Parameter	Value
Architecture	Moirai-2 Small (6 layers, $d = 384$, $d_{\text{ff}} = 1024$, $P = 16$)
Total parameters	11.4M (baseline), 11.4M + 12K (HD10)
Training data	LOTSA v1, Moirai-2 config (weighted, variate-proportional)
Steps	10K (standard), 100K (extended)
Batch size	256
Optimizer	AdamW ($\beta_1 = 0.9$, $\beta_2 = 0.98$, $\epsilon = 10^{-6}$, wd$= 0.1$)
LR schedule	Cosine, peak $\eta = 10^{-3}$, min $\eta = 0$, 1K warmup
Precision	bf16 mixed precision
Seeds	0, 1, 2, 7, 42

Evaluation: GIFT-Eval benchmark [Vijay et al., 2024], 97 configs, context length 4,000, patch size 32 (8 for quarterly). Metric: MASE, aggregated as geometric mean. Statistical test: paired sign test (binomial, two-sided), pooled across 5 seeds (485 comparisons).

B Per-Seed Detailed Results

Table 3: HD10 vs. Baseline: per-seed pairwise comparison (10K steps).

Seed	BL MASE	HD10 MASE	Δ%	Wins/97	p-value
0	0.8666	0.8234	$+4.99\%$	66	2.5×10^{-4}
1	0.8503	0.8465	$+0.44\%$	54	0.155
2	0.8630	0.8391	$+2.77\%$	70	7.4×10^{-6}
7	0.8676	0.8320	$+4.11\%$	78	5.8×10^{-10}
42	0.8614	0.8430	$+2.13\%$	61	7.2×10^{-3}
Pooled			$+2.89\%$	**329/485**	$\mathbf{1.5 \times 10^{-15}}$

C Matched Learning Rate

Table 4: LR $\times$ method interaction (5-seed mean MASE).

	$\eta = 10^{-3}$	$\eta = 2 \times 10^{-3}$	LR sensitivity
Baseline	0.8617	0.8422	2.27%
HD10	0.8368	0.8361	0.09%
HD10 vs BL	-2.89% ($p < 10^{-15}$)	-0.72% ($p = 3.8 \times 10^{-4}$)	

Table 5: Per-seed matched LR comparison ($\eta = 2 \times 10^{-3}$).

Seed	BL@2e-3	HD10@2e-3	Δ%	Wins/97	p-value
0	0.8424	0.8339	$+1.01\%$	57	0.052
1	0.8442	0.8338	$+1.23\%$	55	0.111
2	0.8378	0.8468	-1.07%	46	0.729
7	0.8339	0.8350	-0.13%	54	0.155
42	0.8526	0.8307	$+2.55\%$	68	4.7×10^{-5}
Pooled	**0.8422**	**0.8361**	$+0.72\%$	**280/485**	3.8×10^{-4}

D Complete Ablation Tables

D.1 Polynomial Degree (seed 0, Chebyshev, $s = 16$, 10% dropout)

Degree	Effective lags	MASE	Δ% vs BL	Wins/97
2	1 (at $2P$)	0.8388	-3.20%	63
3	1 (at $3P$)	0.8497	-1.95%	55
4	2 (at $2P$, $4P$)	**0.8234**	-4.99%	**66**
5	2	0.8445	-2.56%	—
6	3 (at $2P$, $4P$, $6P$)	0.8345	-3.71%	69
8	4 (at $2P$–$8P$)	0.8413	-2.92%	62

D.2 Filter Stride (seed 0, Chebyshev $d = 4$, 10% dropout)

Stride	Lookback	MASE	Δ%	Wins/97
16 ($= P$)	32, 64	**0.8234**	-4.99%	**66**
4	8, 16	0.8406	-3.00%	57
8	16, 32	0.8518	-1.71%	62
32	64, 128	0.8484	-2.10%	63

D.3 Filter Basis (seed 0, $d = 4$, $s = 16$, 10% dropout)

Basis	MASE	Δ%	Wins vs BL	vs Cheby
Chebyshev	**0.8234**	-4.99%	66/97	—
EMA	0.8329	-3.89%	65/97	41/97
Legendre $d = 4$	0.8400	-3.07%	62/97	30/97
Finite Diff	0.8569	-1.12%	54/97	29/97
Duplicate	0.8583	-0.96%	42/97	27/97

D.4 Inference-Time Content Ablation (seed 0)

E Multi-Scale Variants

All three methods are statistically indistinguishable head-to-head.

F Context Length

The $\sim$5% benefit is constant across context lengths. HD10 at context 2,000 (0.8331) outperforms baseline at context 4,000 (0.8666).

G Training Dynamics

H Base Model Scale (46M Parameters)

Results are inconsistent at 46M scale, suggesting the benefit is specific to capacity-constrained models.

I FEV-Bench Evaluation

HD10 generalizes beyond GIFT-Eval to the FEV-Bench suite, winning 72 of 100 configurations with consistent improvement in both MASE and squared log metrics.

Inference hint	MASE	vs Normal
Normal (Chebyshev)	0.8234	—
Baseline (no hint)	0.8666	$+5.2\%$
Zeros	0.9003	$+9.3\%$
Random noise	0.9427	$+14.5\%$
Duplicate	1.0699	$+29.9\%$

Table 6: Multi-scale variants (5 seeds, 10K steps).

Method	Mean	Std	vs BL	Wins/485	p
HD10 (d=4, 10% drop)	0.8368	0.0083	-2.89%	329	1.5×10^{-15}
MS46 (d=4+6, 0% drop)	0.8434	0.0123	-2.13%	330	6.8×10^{-16}
MSHD10 (d=4+6, 10% drop)	0.8437	0.0072	-2.09%	315	2.2×10^{-11}

J Matched-Official Training Protocol (100K Steps)

Under the official Moirai-2 training protocol (10K warmup steps, 100K total cosine schedule), HD10 still provides a 2.3% improvement. The baseline's seed 7 (0.917) shows the late instability noted in §5.5, which HD10 avoids (0.880).

K Leaderboard Context

HD10 at 11.4M parameters achieves 0.828 normalized MASE, surpassing Moirai 1.0 Large (311M, 0.875) with $27\times$ fewer parameters.

Context	BL	HD10	Δ%	Wins/97
512	0.9432	0.8986	−4.73%	62
1,000	0.9166	0.8706	−5.02%	65
2,000	0.8813	0.8331	−5.47%	71
4,000	0.8666	0.8234	−4.99%	66

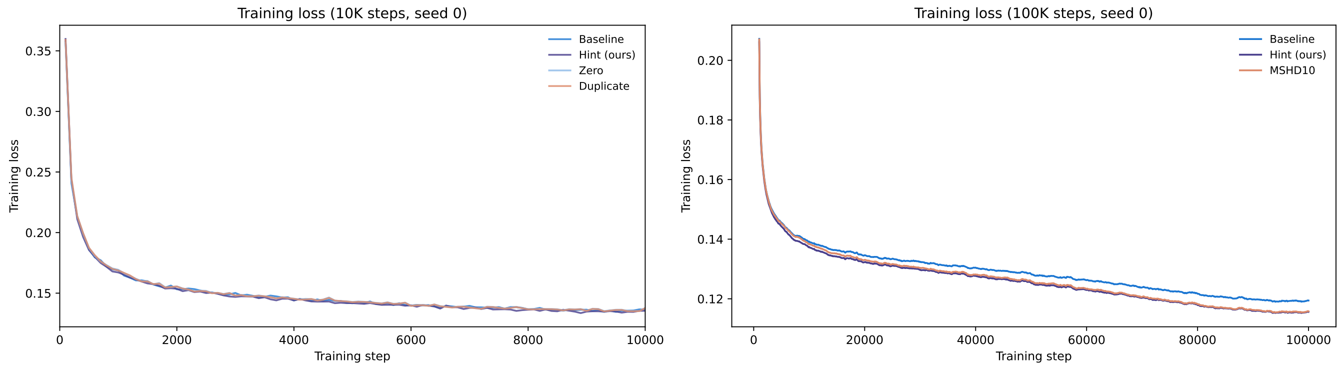

Figure 5: Training loss at 10K (left) and 100K (right) steps. HD10 maintains lower loss throughout. The baseline's eval collapse at 80K is not reflected in training loss.

Table 7: Training loss means (5 seeds) vs. eval MASE. Training loss differences are small (1–2.4%) while eval improvements are much larger (2.9–3.9%), confirming hints act as a regularizer rather than simply improving optimization.

	10K steps		100K steps	
Method	**Train loss**	**Eval MASE**	**Train loss**	**Eval MASE**
Baseline	.1364	0.862	.1187	0.878
HD10 (ours)	**.1350**	**0.837**	**.1159**	**0.844**
Zero	.1366	0.858	.1187	0.894
Duplicate	.1366	0.865	.1190	0.886
MSHD10	—	—	.1146	0.838†
HD10 Δ% vs BL	−1.0%	−2.9%	−2.4%	−3.9%
Amplification	2.9×		1.6×	

†MSHD10 eval mean over 3 available seeds (0, 2, 7).

Seed	BL (46M)	HD10 (46M)	Δ%	Wins
0	0.8504	0.8520	−0.19%	52/95
1	0.8473	0.8509	−0.42%	60/95
42	0.8570	0.8417	+1.80%	59/93

Table 8: FEV-Bench results (100 task configurations). Geo = geometric mean MASE. SQL = squared log metric.

Method	MASE (geo)	SQL (geo)	Wins/100	Δ% MASE
HD10 (Ours)	**1.253**	**1.657**	**72/100**	+2.5%
Baseline	1.284	1.679	—	—

Table 9: Matched-official protocol: 10K warmup, 100K cosine schedule (matching the official Moirai-2 training configuration). Normalized MASE reported.

Method	s0	s2	s7	Mean	Δ%
BL (10K warm)	0.882	0.880	0.917	0.893	—
HD10 (10K warm)	**0.892**	**0.846**	**0.880**	**0.873**	+2.3%

Table 10: GIFT-Eval leaderboard context: normalized MASE (lower is better). HD10 Small (11.4M) outperforms Moirai 1.0 Large (311M) with 27× fewer parameters.

Model	Parameters	Normalized MASE
HD10 Small (Ours)	**11.4M**	**0.828**
Moirai-2 Small (retrained BL)	11.4M	0.862
Moirai 1.0 Large	311M	0.875